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The perceptual motivation for the Norwegian merger of /ʃ/ and /ç/

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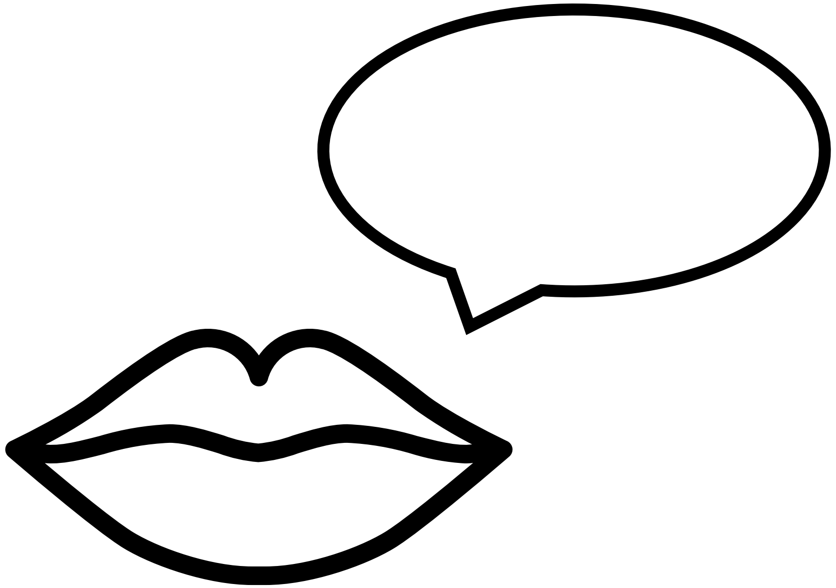
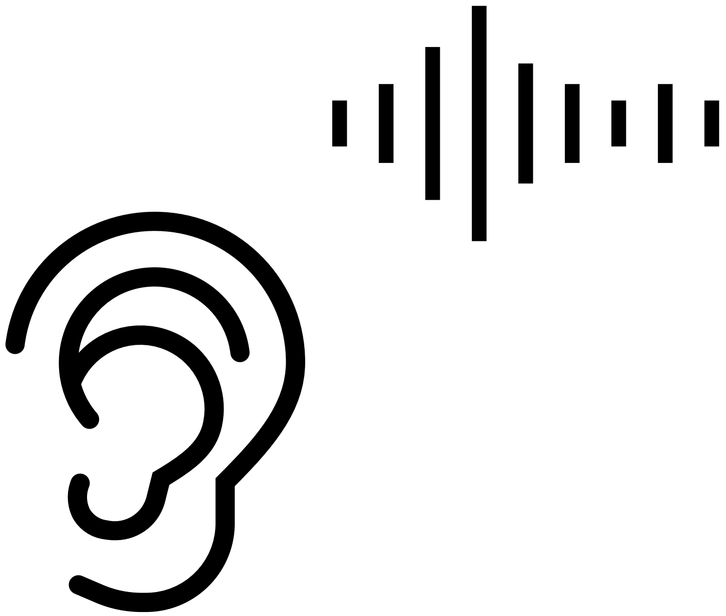
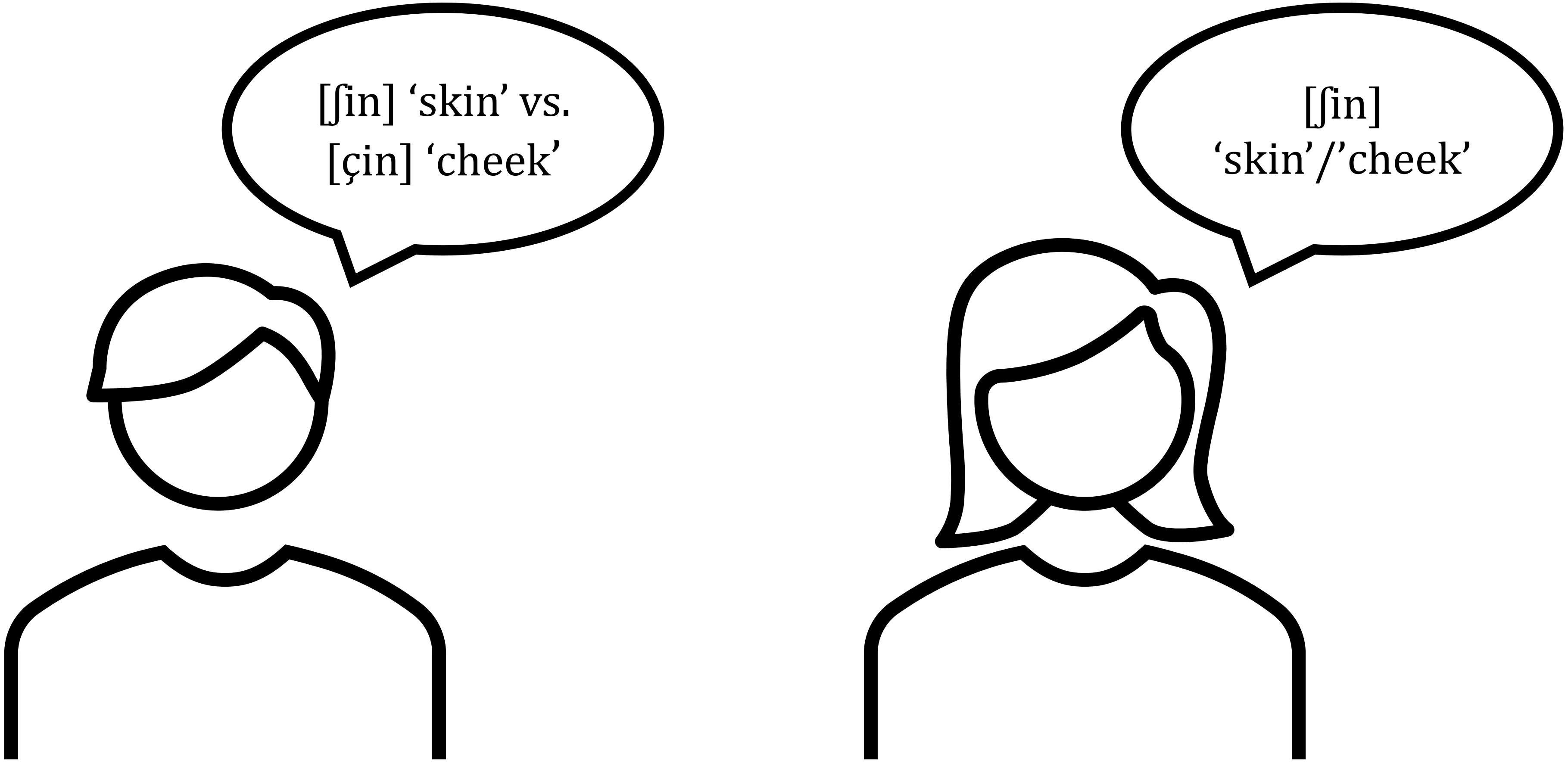
What motivates sound change?





Paul (1920); Martinet (1964); Ohala (1981); Blevins (2004).

/ʃ/ and /ç/ in Norwegian



The merger of /ʃ/ and /ç/

Found in various cities in Norway.

Descriptions of frequency and distribution.

No studies investigating the phonetic motivation.

E.g. Andrésen (1980); Johannessen (1983); Hegland (1992); Papazian (1994); Dalbakken (1996).

[ç]: a difficult sound to produce?

Child language acquisition

Last phoneme acquired by children.

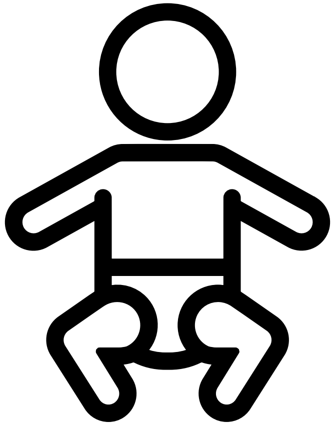
Cross-linguistic frequency

Rare sound cross-linguistically.

Articulation

Dorsal /ç/ more difficult than coronal /ʃ/.

Papazian (1994); Torp (1999); van Dommelen (2003).



Child language acquisition

Vanvik (1971): one child speaker.

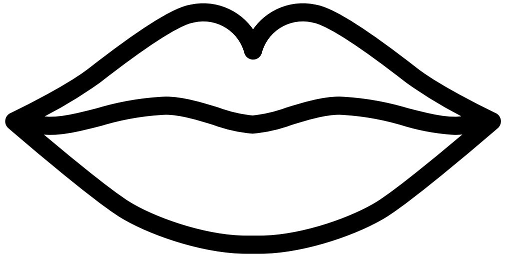
Fintoft et al. (1983): /ʃ ç s r/ all acquired late.

Simonsen (1990): high number of mistakes for both /ʃ/ and /ç/.

Cross-linguistic frequency

Maddieson (1984): rare phoneme.

Roach (2009) and Okada (1999): [ç] found as allophone in English and Japanese.



Articulation

Simonsen and Moen (2004): /ç/ is coronal for most speakers.

Evjen and Stausland (2025): outcome of the merger can be [ç].

Avoidance of [ç] cannot explain merging with [ʃ] in particular.

An alternative explanation

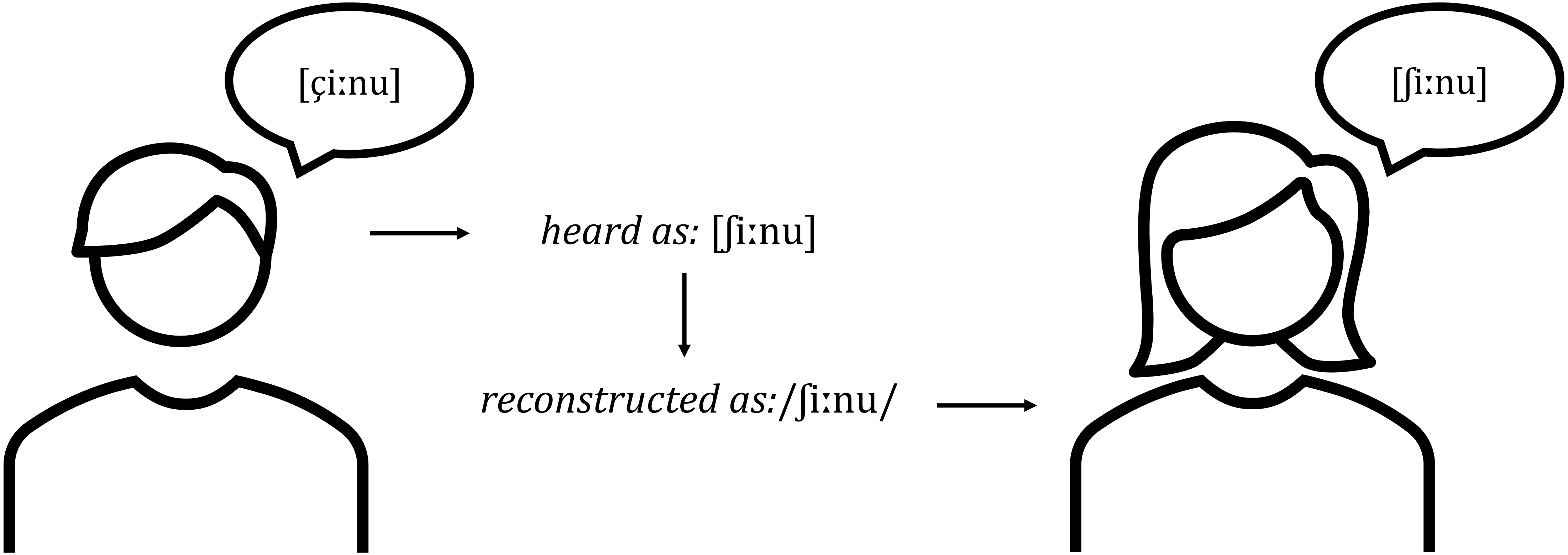
Perceptual similarity

Ohala (1981): listeners’ misperception of noisy signals is a common source of sound change.

Blevins (2004): merger of /f/ and /θ/ in some English dialects.

Stausland Johnsen (2012): merger of apical alvolars and apical postalveolars in many Norwegian dialects.

Perceptual similarity



Ohala (1981).

The current study

Perceputal distance between /f s ʃ ç/.

Confusability of /ʃ/ and /ç/ compared to other pairs.

Smaller perceptual distance increases chance of misperceptions and subsequently merger.

Luce (1963); Hautus et al. (2022).

Methods

59 native Norwegian participants (45 female, mean age 24).

4 native Norwegian speakers (3 female, mean age 25).

24 CV nonce words per speaker (/f s ʃ ç/ × /i ɑ u/ × 2 repetitions).

AX discrimination experiment.

Evjen et al. (2025).

Analysis

Six pairs: ʃ – ç, ç – f, ç – s, ʃ – f, ʃ – s, f – s.

Measures:

Proportions correct: rate of correct ‘different’ and ‘same’ trials.

d’ (d-prime): Z(Hit rate) – Z(False alarm rate).

Hautus et al. (2022).

Analysis

Statistical analyses:

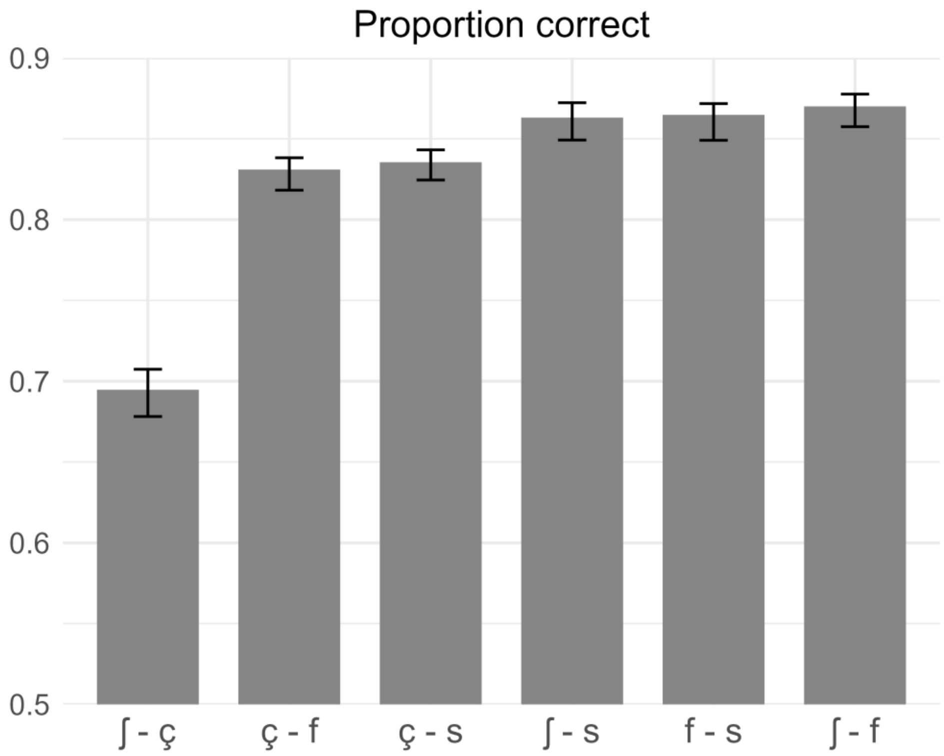
Multidimensional scaling (MDS): visual representation of perceptual distances as indicated by the *d'* values.

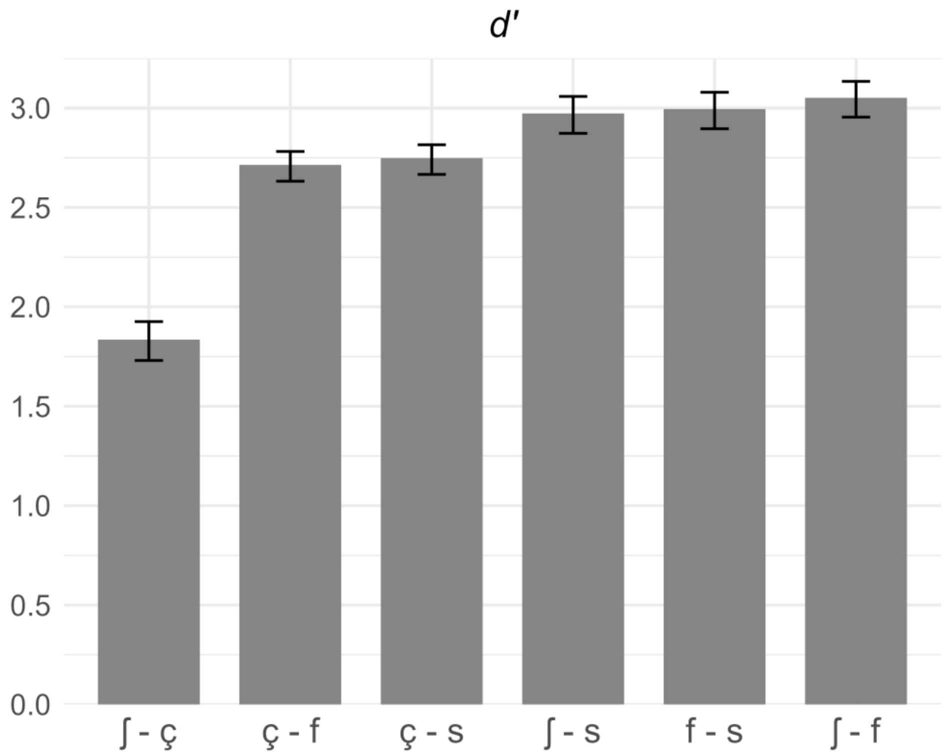
Bayesian mixed effects model: approximate posterior distributions of plausible parameter values, given the data, the model, and the priors.

$d' \sim$ pair + (1 + pair | listener) + (1 + pair | speaker)

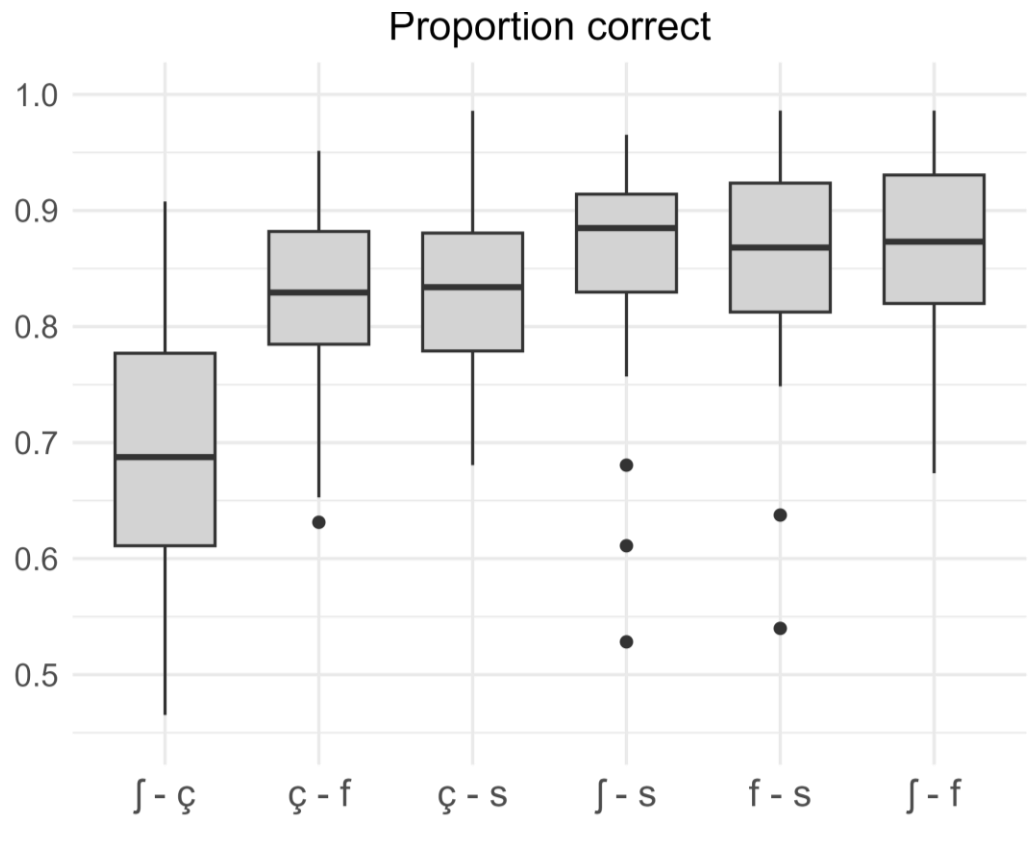
Kruskal & Wish (1978); Cox & Cox (2001).

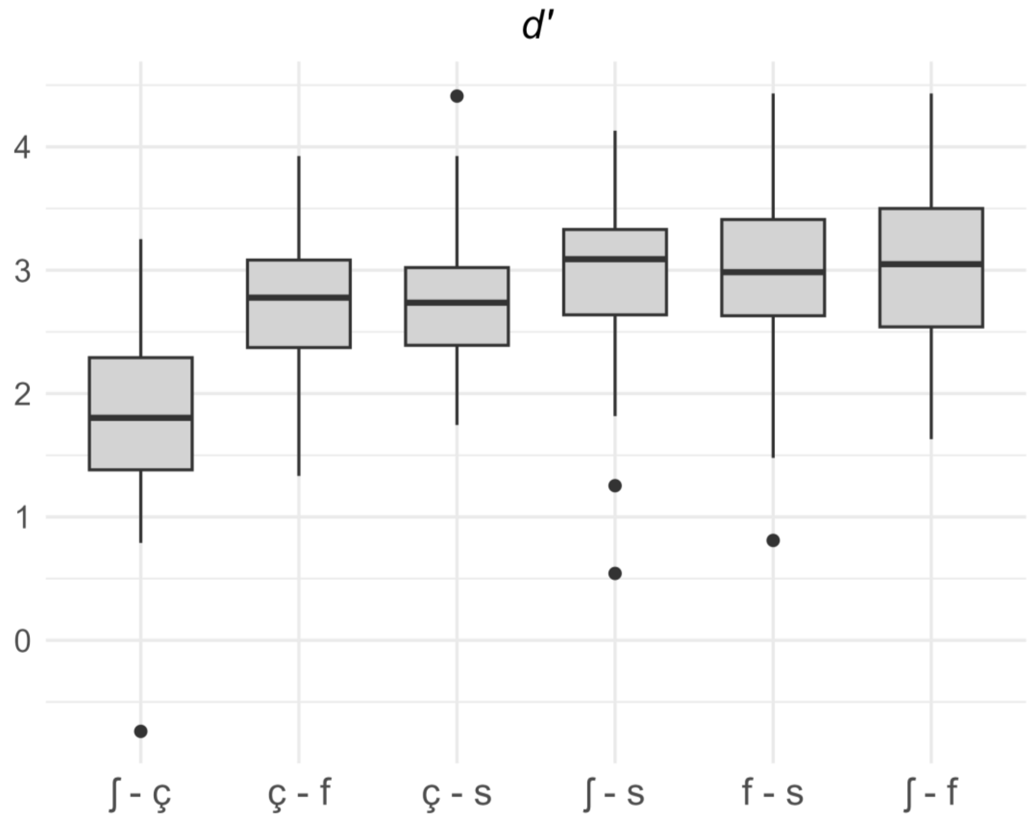
Descriptive results



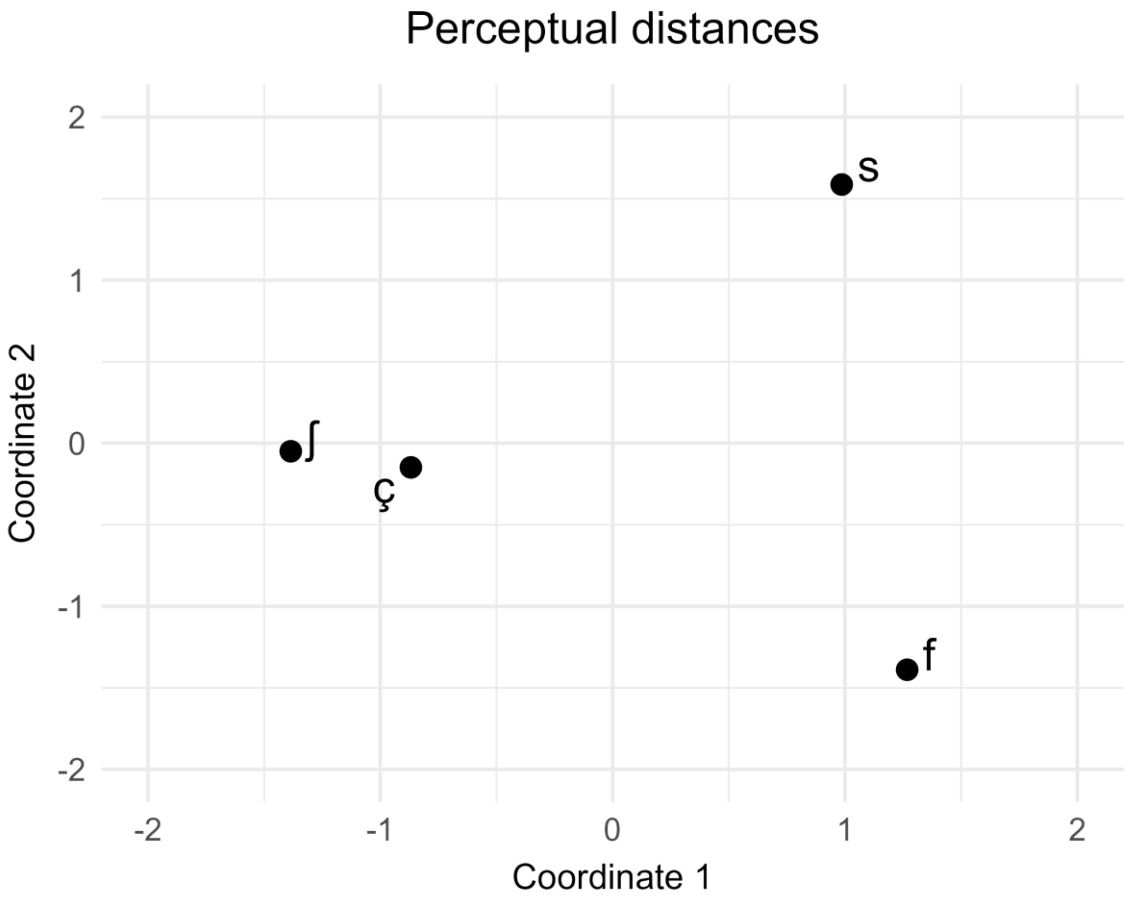


Descriptive results



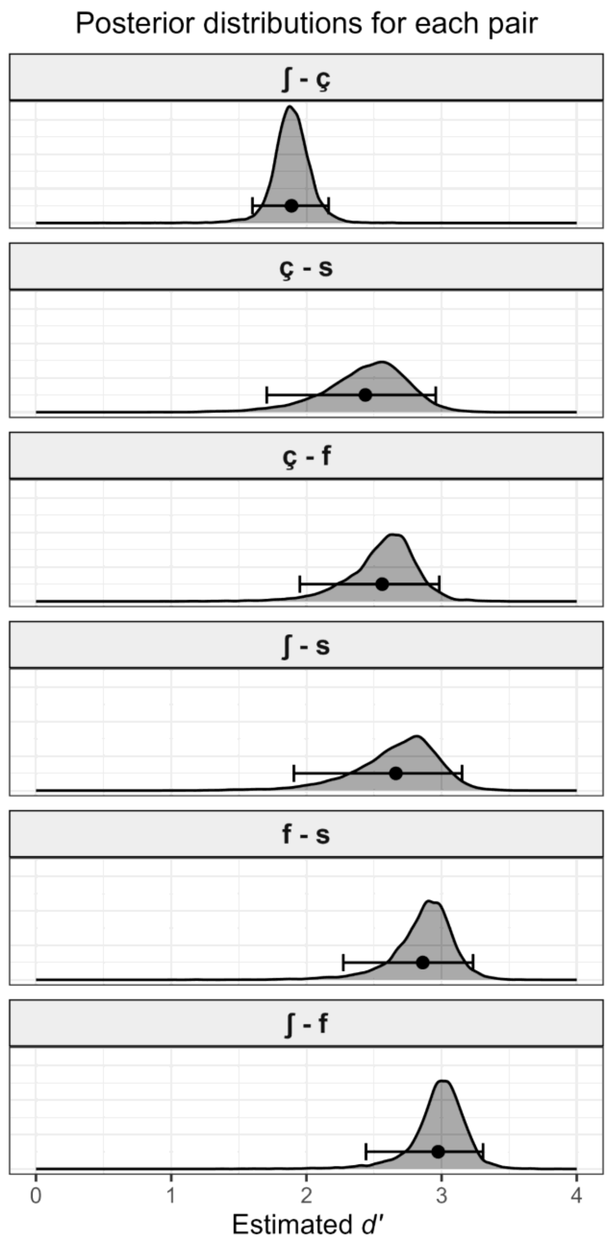


Multidimentional scaling



Regression model

Pair	β	LCI	HCI	Pr(β > 0)
ʃ - ç	1.89	1.60	2.16	1
ç - s	0.55	-0.12	1.03	0.953
ç - f	0.67	0.10	1.03	0.986
ʃ - s	0.77	0.05	1.22	0.980
f - s	0.97	0.44	1.26	0.996
ʃ - f	1.08	0.63	1.33	0.998



Summing up

/ʃ/ and /ç/ were confused more often than other pairs.

Smaller perceputal distance between /ʃ/ and /ç/.

Listener misperceptions as explanation for the merger of /ʃ/ and /ç/.

Summing up

First study investigating the motivation for the merger of /ʃ/ and /ç/.

Findings align with observed outcomes of the merger (Evjen and Stausland, 2025).

Important role of perception in driving forward sound change.

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Summing up

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